CURRENT RESEARCH / COMPUTATIONAL PROTEIN DESIGN

# AI-designed bispecific binders for Alzheimer's disease

Aβ42 targeting × transferrin receptor 1 (TfR1) | Goal: receptor-mediated delivery across the BBB

**01 TARGET PREPARATION**
**Structure-guided epitopes**
Aβ42: 9CO4 • TfR1: 6WRV

**02 DE NOVO DESIGN**
**BindCraft + ProteinMPNN**
Generate, redesign, AF2-score

**03 FILTER & RANK**
**Stability + developability**
23 Aβ42 / 224 TfR1 retained

**04 FUSION LIBRARY**
**5 Aβ42 × 5 TfR1 binders**
5 linkers × 2 domain orders

**05 FOLD & SCREEN**
**AlphaFold2 / ColabFold**
250 fusion predictions scored

**442**
**accepted arm designs**
62 Aβ42 + 380 TfR1[1]

**250**
**fusion designs modeled**
25 binder pairs × 10 variants

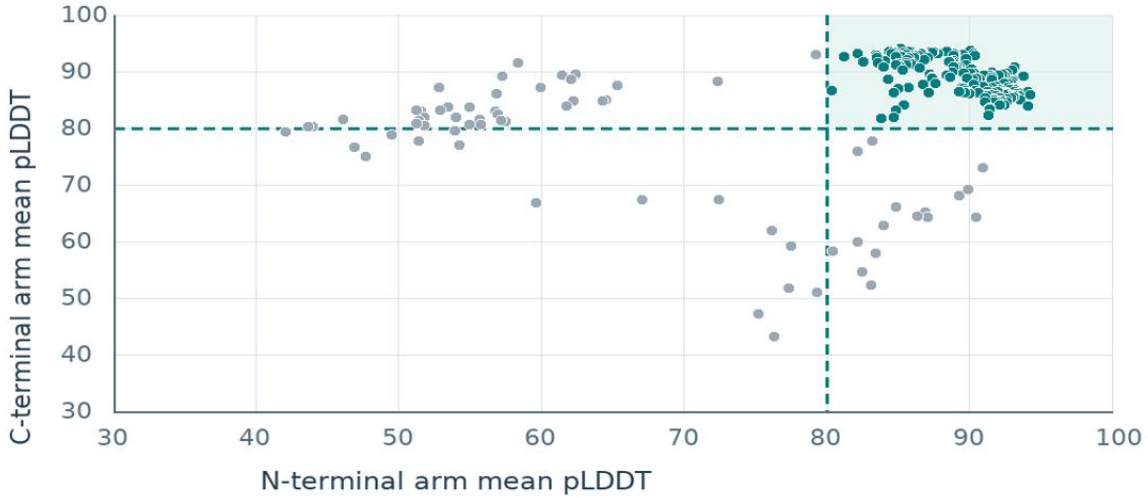
**183 / 250**
**pass Stage 8 gates**
73.2% computational pass rate

**FUSION-LEVEL COMPARISON • All 250 designs; one top-ranked prediction per design**

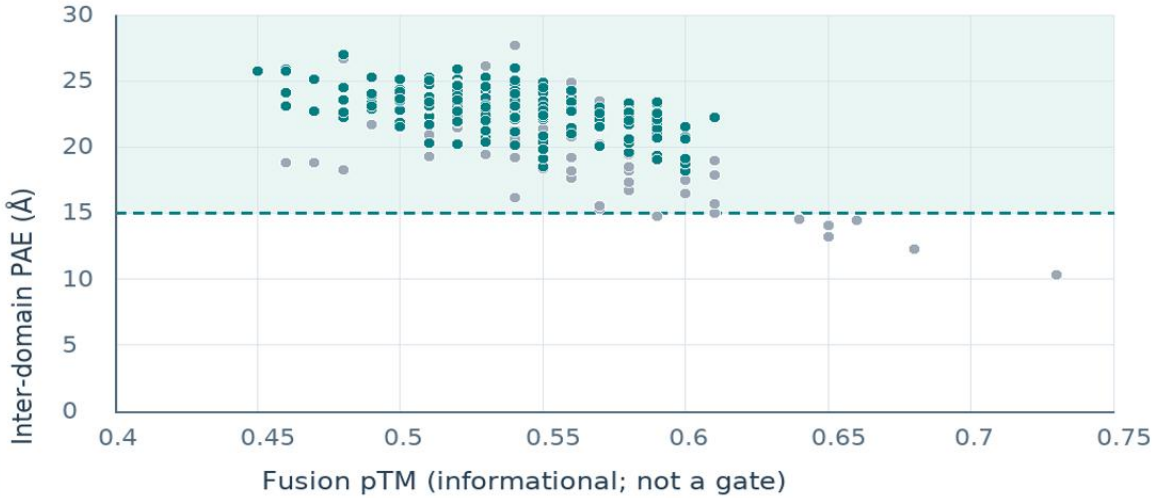
● Pass all gates (n = 183) ● Fail ≥1 gate (n = 67)

Dashed lines: required thresholds

## Per-arm folding confidence across fusion designs




## Whole-chain confidence vs. domain independence




Gate: both arms pLDDT >80 AND inter-domain PAE >15 Å. High inter-domain PAE reflects uncertain relative orientation.

Computational candidates; binding, selectivity and BBB transport remain unvalidated. Fusion ipTM was not calculated.

[1] Campaign-reported library total. Separate RFdiffusion → ProteinMPNN route documented; not the source of this fusion cohort.